
Human-Assisted Learning for OOD Generalization and Detection with Diverse Sample Selection

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Abstract

Modern machine learning systems frequently encounter heterogeneous distribution shifts during deployment, including both covariate shifts and semantic out-of-distribution (OOD) samples. While prior work has largely treated OOD generalization and OOD detection as separate problems, real-world scenarios require handling both simultaneously. This work studies a human-assisted learning framework in which a limited labeling budget is used to selectively annotate informative samples from a mixed wild data distribution. Several sample selection strategies are investigated, including Top-k sampling, ambiguity-based selection (AHA), and a diversity-enhanced extension (AHA-Diverse). Experimental results on CIFAR-10 with multiple OOD datasets demonstrate that ambiguity-based selection improves robustness by refining decision boundaries, while diversity-aware selection improves representation coverage. The results highlight a fundamental tradeoff between classification accuracy and OOD detection performance, and show that combining uncertainty-based selection with diversity leads to more balanced performance under heterogeneous distribution shifts.

1 Problem Statement

Modern machine learning systems deployed in real-world environments frequently encounter distribution shifts that degrade performance and reliability. These shifts arise in two primary forms: covariate out-of-distribution (OOD) shifts, where the input distribution changes while the label space remains fixed, and semantic OOD shifts, where entirely new, unseen classes appear at test time. Existing research typically addresses these challenges independently—OOD generalization methods aim to maintain classification accuracy under covariate shift, while OOD detection methods focus on identifying and rejecting semantically novel inputs. However, real-world deployment scenarios naturally involve a mixture of in-distribution data, covariate-shifted data, and semantically novel data, creating a unified and more challenging learning problem.

In this scenario, the learner has a limited budget for human labeling but has access to a vast pool of unlabeled "wild" data derived from this heterogeneous mixture as well as labeled in-distribution training data. Therefore, the main issue is how a learning system may intentionally choose a small fraction of wild samples for annotation in a way that improves OOD detection and generalization performance at the same time. Finding informative areas in the model's uncertainty or score space where human feedback provides the most value is necessary to solve this issue. The objective is to create a human-assisted, label-efficient framework that efficiently makes use of sparse annotations to improve resilience to deployment-related heterogeneous distribution adjustments.

34 2 Problem Setup

35 We consider a supervised learning setting where a classification model is deployed in an envi-
 36 ronment exhibiting heterogeneous distribution shifts. Let $\mathcal{X} \subset \mathbb{R}^d$ denote the input space and
 37 $\mathcal{Y} = \{1, \dots, K\}$ denote the label space consisting of K known classes.

38 We are given an initial labeled in-distribution dataset:

$$S_{in} = \{(x_i, y_i)\}_{i=1}^n \sim P_{XY}^{in}.$$

39 During deployment, the model encounters samples from:

$$P_{wild} = (1 - \pi_c - \pi_s)P_{in} + \pi_c P_{out}^{cov} + \pi_s P_{out}^{sem}.$$

40 A limited labeling budget B allows querying:

$$S_{human} = \{(x_t, \tilde{y}_t)\}_{t=1}^B.$$

41 An OOD detector is defined as:

$$D_{\theta}(x; \lambda) = \begin{cases} \text{ID}, & g_{\theta}(x) > \lambda \\ \text{OOD}, & g_{\theta}(x) \leq \lambda \end{cases}$$

42 The objective is to select samples that maximize classification accuracy while minimizing detection
 43 error.

44 3 Related Work

45 Out-of-distribution (OOD) learning has been widely studied in two separate directions: OOD gen-
 46 eralization and OOD detection. OOD generalization focuses on maintaining classification perfor-
 47 mance under covariate shifts, where the input distribution changes but the label space remains fixed.
 48 Prior work in this area includes invariant risk minimization approaches (2), domain generalization
 49 techniques (14), and representation learning methods designed to extract domain-invariant features.
 50 These approaches aim to improve robustness by learning features that generalize across different
 51 environments.

52 In contrast, OOD detection focuses on identifying samples that do not belong to the training distri-
 53 bution. A variety of methods have been proposed, including confidence-based approaches, energy-
 54 based scoring (19), and distance-based methods such as Mahalanobis scoring. While these methods
 55 are effective for detecting semantic OOD samples, they often do not address classification perfor-
 56 mance under covariate shifts.

57 More recent work has attempted to combine these two perspectives by leveraging unlabeled data
 58 from a wild distribution. Methods such as SCONE (7) utilize mixed distributions containing both in-
 59 distribution and OOD samples to improve robustness. Additionally, active learning methods (4) and
 60 diversity-based sampling strategies have been explored to improve sample efficiency when labeling
 61 budgets are limited.

62 However, most existing approaches either focus on detection or generalization independently, or rely
 63 on passive use of unlabeled data. The framework considered in this work builds upon these ideas
 64 by introducing a human-assisted selection strategy that actively identifies informative regions in the
 65 data distribution, combining uncertainty-based and diversity-based sampling to improve both OOD
 66 generalization and detection simultaneously.

67 4 Method

68 A human-assisted learning framework is considered to jointly address out-of-distribution (OOD)
 69 generalization and OOD detection under heterogeneous distribution shifts. In contrast to standard
 70 supervised learning, where models are trained solely on labeled in-distribution data, the framework
 71 assumes access to a large pool of unlabeled wild samples. A limited labeling budget is used to
 72 query informative examples from this pool, which are then incorporated into the training process.
 73 The central challenge is to determine which samples should be selected such that both classification
 74 accuracy and detection performance improve simultaneously.

75 **OOD Scoring.** To guide sample selection, an energy-based scoring function is used to measure
 76 the likelihood of a sample being out-of-distribution. Given a classifier $f_w(x) \in R^K$, the energy
 77 score is defined as

$$E(x) = -\log \sum_{k=1}^K \exp(f_w^{(k)}(x)).$$

78 This score provides a continuous ordering of samples from likely in-distribution to likely OOD.
 79 Compared to confidence-based measures, the energy score is directly related to the model’s logit
 80 structure and has been shown to correlate well with data density, making it suitable for both ranking
 81 and threshold-based detection.

82 **Learning Framework.** The model is trained in two stages. First, a base classifier is trained on
 83 the labeled in-distribution dataset S_{in} . Next, a subset of samples S_{human} is selected from the wild
 84 pool S_{wild} and labeled using the available annotation budget. The model is then finetuned on the
 85 combined dataset $S_{in} \cup S_{human}$. This allows the model to incorporate information from both in-
 86 distribution and OOD samples, improving robustness.

87 **Top-k Selection.** The simplest selection strategy is Top-k sampling, where the k samples with the
 88 highest energy scores are selected. This approach prioritizes extreme OOD examples, which are
 89 highly informative for improving classification boundaries. However, it does not explicitly consider
 90 uncertainty or ambiguity in the model’s predictions, and may overemphasize easily identifiable OOD
 91 samples while neglecting more informative boundary cases.

92 **AHA Selection.** To address this limitation, the AHA strategy focuses on selecting samples from
 93 an ambiguity region in score space. Instead of choosing extreme samples, AHA identifies a region
 94 around a threshold where the model is uncertain whether a sample is in-distribution or OOD. For-
 95 mally, this corresponds to selecting samples near a decision boundary defined by the OOD score.
 96 These samples provide maximal information for refining both classification and detection bound-
 97 aries, as they represent regions where the model is most likely to make mistakes.

98 **AHA-Diverse Selection.** While AHA focuses on uncertainty, it may still select redundant samples
 99 that are similar in feature space. To improve coverage, a diversity-aware extension is introduced.
 100 In AHA-Diverse, feature representations extracted from the model are clustered using KMeans,
 101 and samples are selected from different clusters within the ambiguity region. This ensures that
 102 the labeled set contains diverse examples spanning multiple regions of the input space, reducing
 103 redundancy and improving generalization.

104 **Intuition and Connection to Prior Work.** The proposed framework is motivated by the idea that
 105 labeling samples near overlapping regions of distributions provides the highest information gain.
 106 In heterogeneous wild data, covariate OOD and semantic OOD distributions may overlap in score
 107 space. Selecting samples from these regions allows the model to better distinguish between different
 108 types of shifts. This intuition is closely related to the concept of a maximum disambiguation region,
 109 where labeling samples leads to the largest improvement in both OOD detection and generalization.

110 **Training Objective.** The final model jointly optimizes classification and detection objectives. The
 111 classification component is trained using standard cross-entropy loss on labeled in-distribution and
 112 covariate OOD samples, while the detection component learns to separate semantic OOD samples
 113 from in-distribution data. This combined objective ensures that the model maintains high accuracy
 114 while improving robustness to distribution shifts.

115 5 Experiments

116 A comprehensive experimental framework is designed to evaluate the effectiveness of human-
 117 assisted sample selection strategies under heterogeneous distribution shifts. The objective is to
 118 simultaneously assess classification performance and OOD detection capability in a setting that
 119 closely resembles real-world deployment scenarios.

120 The in-distribution dataset used throughout the experiments is CIFAR-10, which consists of natural
 121 images from ten object categories. To simulate realistic distribution shifts, a heterogeneous wild

dataset is constructed by combining three types of data sources. First, in-distribution samples are included to reflect the presence of familiar data during deployment. Second, covariate OOD samples are generated using CIFAR-10-C, which introduces a variety of common corruptions such as Gaussian noise, blur, contrast variation, and brightness changes. These samples retain the same label space as the original dataset but exhibit significant shifts in appearance. Third, semantic OOD samples are drawn from external datasets including SVHN, Textures, and Places365. These datasets introduce entirely new semantic classes, ranging from digit images to texture patterns and complex scene categories, thereby increasing the difficulty of the detection task.

The wild data distribution is therefore a mixture of in-distribution, covariate-shifted, and semantically novel samples, reflecting the heterogeneous nature of real-world environments. Importantly, the learner does not have access to labels for these samples and must rely on a limited annotation budget to select the most informative examples.

A WideResNet-40-2 architecture is used as the base model due to its strong performance on CIFAR benchmarks. The model is first trained on CIFAR-10 for 100 epochs using stochastic gradient descent with Nesterov momentum and cosine learning rate decay. This initial training phase establishes a baseline representation of in-distribution data. Following this, a subset of samples is selected from the wild dataset using a specified selection strategy. These selected samples are then labeled and incorporated into a finetuning stage, where the model is retrained using both the original labeled data and the newly annotated samples.

The selection strategies considered in this study represent different approaches to utilizing the labeling budget. Top-k sampling selects samples with the highest OOD scores, effectively prioritizing extreme cases that are most dissimilar from the training data. AHA focuses on samples near an ambiguity region in the score space, where the model is uncertain whether inputs belong to the in-distribution or OOD. AHA-Diverse extends this approach by enforcing diversity in the selected samples through clustering in feature space, ensuring that the labeled set covers a broader portion of the input distribution.

Performance is evaluated using four complementary metrics. In-distribution accuracy measures classification performance on CIFAR-10 test data. Covariate OOD accuracy evaluates performance on corrupted CIFAR-10-C data, reflecting generalization under domain shifts. The false positive rate at 95% true positive rate (FPR95) measures the frequency with which OOD samples are incorrectly classified as in-distribution, capturing detection reliability. Finally, AUROC provides an overall measure of the model’s ability to distinguish between in-distribution and OOD samples across different thresholds.

Together, these metrics provide a comprehensive view of how different selection strategies affect both classification and detection performance, enabling a detailed comparison of their strengths and limitations.

6 Results

6.1 Quantitative Results

The performance of the evaluated selection strategies across multiple datasets is summarized in Table 1. The results demonstrate clear differences in behavior between methods when applied to heterogeneous distribution shifts. While all methods achieve comparable in-distribution accuracy, their performance diverges significantly when evaluated on out-of-distribution data, particularly in terms of detection metrics such as FPR95 and AUROC.

For the SVHN dataset, all methods perform strongly due to the clear separation between in-distribution object classes and digit images. In this relatively simple setting, Top-k achieves slightly higher OOD accuracy, indicating that selecting extreme samples is sufficient when the distribution gap is large. However, AHA-Diverse achieves the lowest FPR95, suggesting that incorporating diversity improves detection reliability even in easier scenarios.

The Textures dataset presents a more challenging setting, where the distinction between in-distribution and OOD samples is less obvious. In this case, Top-k achieves higher classification accuracy, but AHA-Diverse provides improved AUROC, indicating better ranking of OOD samples. AHA, which focuses on ambiguous regions, performs slightly worse in classification but demon-

Dataset	Method	ID Acc	OOD Acc	FPR95	AUROC
SVHN	Top-k	86.68	79.60	1.33	99.75
SVHN	AHA	86.25	78.33	1.54	99.67
SVHN	AHA-Diverse	86.50	78.28	0.61	99.86
Textures	Top-k	85.98	80.24	28.30	92.59
Textures	AHA	85.33	77.16	31.28	92.03
Textures	AHA-Diverse	85.57	78.80	28.46	92.71
Places365	Top-k	84.77	78.14	72.89	81.46
Places365	AHA	85.84	77.35	60.83	85.89
Places365	AHA-Diverse	85.31	78.24	63.03	84.78

Table 1: Performance comparison across multiple OOD datasets.

Method	ID Acc	OOD Acc	FPR95	AUROC
AHA	85.81	77.61	31.22	92.53
AHA-Diverse	85.79	78.44	30.70	92.45
Top-k	85.81	79.33	34.17	91.27

Table 2: Average performance across all datasets.

strates more consistent behavior in detection metrics. This suggests that ambiguity-based selection begins to play a more important role as the distribution shift becomes more subtle. The most challenging dataset, Places365, highlights the limitations of extreme sample selection. Top-k exhibits significantly higher FPR95, indicating poor detection performance under complex semantic shifts. In contrast, AHA substantially reduces FPR95 by focusing on samples near the decision boundary, where the model is most uncertain. AHA-Diverse maintains competitive performance, balancing classification accuracy and detection, although at the cost of slightly increased false positives compared to AHA.

6.2 Overall Trends

Across all datasets, a consistent pattern emerges in the behavior of the selection strategies. Top-k sampling prioritizes extreme OOD examples, which leads to strong classification performance but fails to generalize well under more complex distribution shifts. AHA improves robustness by focusing on ambiguous samples that lie near the boundary between in-distribution and OOD regions. AHA-Diverse further enhances this approach by ensuring that selected samples are not only informative but also diverse, resulting in improved representation coverage.

This tradeoff between accuracy and detection is evident in the aggregated results. While Top-k achieves strong OOD accuracy, it suffers from higher FPR95, particularly in difficult settings. AHA achieves the lowest FPR95, indicating better detection performance, while AHA-Diverse provides a balanced compromise between the two objectives.

6.3 Score Distribution Analysis

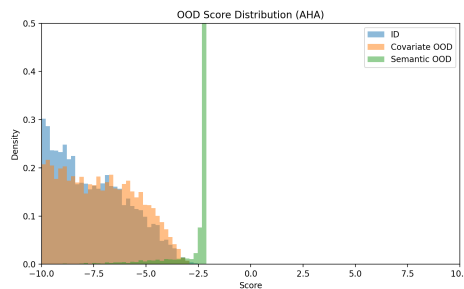


Figure 1: Energy score distribution across ID, covariate OOD, and semantic OOD samples

194 The energy score distribution provides insight into the underlying difficulty of the problem. Semantic
 195 OOD samples occupy a clearly separated region in score space, making them relatively easy to
 196 detect. In contrast, covariate OOD samples overlap significantly with in-distribution data, which
 197 explains the difficulty in distinguishing between these two categories. This overlap directly con-
 198 tributes to higher false positive rates in more challenging scenarios and highlights the importance of
 199 selecting informative samples near overlapping regions.

200 6.4 Qualitative Analysis

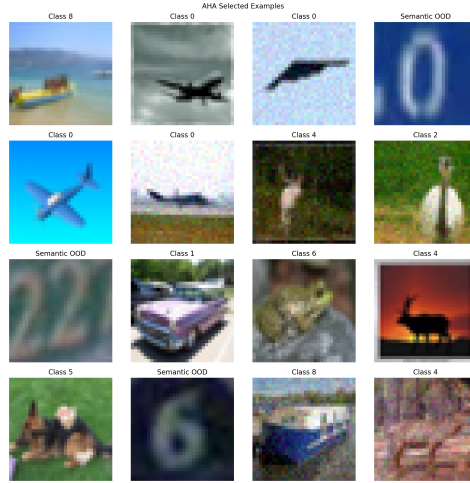


Figure 2: Samples selected by AHA

201 **AHA Sample Selection.** The samples selected by AHA, shown in Figure 2, are concentrated
 202 around regions of ambiguity in the input space. Many of these samples exhibit visual characteristics
 203 that make them difficult to classify confidently, such as partially corrupted images, unusual textures,
 204 or patterns that do not clearly belong to any known class. This behavior is consistent with the design
 205 of AHA, which prioritizes samples near the decision boundary where the model is most uncertain.
 206 By focusing on such ambiguous samples, the model receives informative feedback that helps refine
 207 both classification and detection boundaries. However, it can be observed that several selected
 208 samples share similar visual properties, indicating that AHA may repeatedly select points from a
 209 narrow region of the feature space.

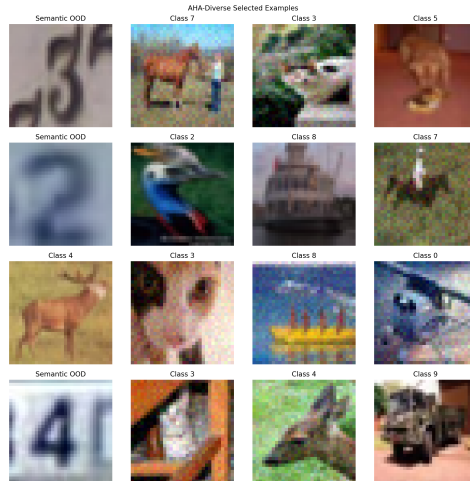


Figure 3: Samples selected by AHA-Diverse

AHA-Diverse Sample Selection. The samples selected by AHA-Diverse, shown in Figure 3, exhibit a broader range of visual patterns compared to AHA. In addition to including ambiguous samples near decision boundaries, AHA-Diverse captures examples from multiple distinct regions of the input space. This includes a mixture of corrupted images, texture patterns, and semantically different structures, reflecting improved diversity in the selection process. The inclusion of clustering in feature space encourages coverage of different modes of the data distribution, reducing redundancy and providing more comprehensive training signals. As a result, AHA-Diverse promotes better representation learning by exposing the model to a wider variety of challenging examples.

Comparison Between AHA and AHA-Diverse. A direct comparison between the two methods reveals a clear tradeoff between focus and coverage. AHA concentrates on highly ambiguous samples, which are effective for refining decision boundaries but may lack diversity. In contrast, AHA-Diverse sacrifices some level of precision in selecting purely boundary samples in order to improve coverage across the feature space. This difference is reflected in the experimental results, where AHA achieves stronger detection performance in some cases, while AHA-Diverse provides more balanced performance through improved generalization. These observations highlight the importance of combining uncertainty and diversity when selecting samples for human-assisted learning.

6.5 Feature Space Visualization

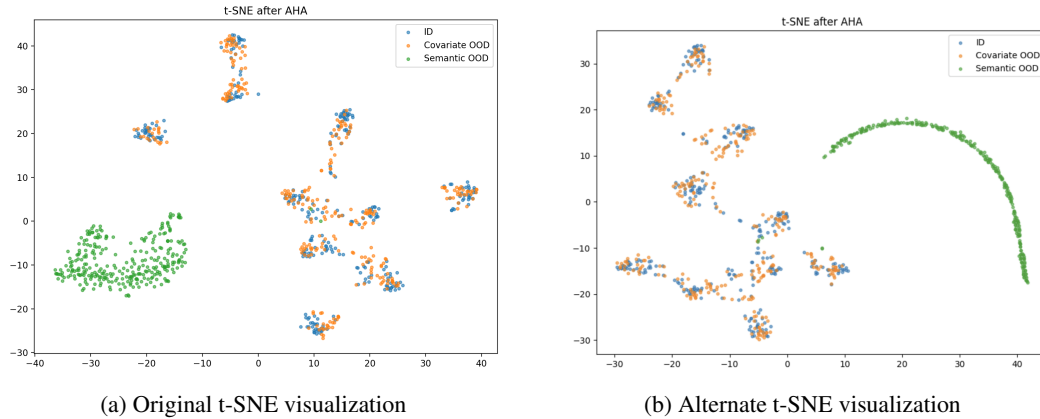


Figure 4: Comparison of t-SNE feature representations across runs. While the geometric structure differs due to the stochastic nature of t-SNE, semantic OOD samples consistently form a distinct region separated from in-distribution data.

The t-SNE visualization illustrates the structure of the learned feature space. Semantic OOD samples form a distinct region that is clearly separated from in-distribution data, while covariate OOD samples remain closer to in-distribution clusters. This behavior reflects the inherent similarity between covariate shifts and original data. Although the geometric structure appears curved due to the stochastic and non-linear nature of t-SNE, the key observation is the consistent separation of semantic OOD samples, which indicates effective feature learning.

Although the two visualizations exhibit different geometric structures, both consistently demonstrate separation between semantic OOD and in-distribution samples. The variation in shape is expected due to the stochastic and non-linear nature of t-SNE embeddings and does not affect the underlying conclusion regarding feature separation.

7 Limitations

While the proposed framework demonstrates strong performance across multiple datasets, several limitations should be noted. First, the implementation relies on an approximate thresholding strategy for identifying the ambiguity region, rather than the more precise noisy binary search formulation described in prior work. As a result, the selected samples may not perfectly align with the theoretically optimal disambiguation region.

243 Second, the experiments are conducted on relatively small-scale datasets such as CIFAR-10 and its
244 variants. Although these datasets are commonly used for benchmarking, they may not fully capture
245 the complexity of real-world deployment scenarios. Performance on larger-scale datasets or more
246 diverse domains may exhibit different behavior.

247 Third, the effectiveness of AHA-Diverse depends on the quality of feature representations used for
248 clustering. If the learned features are not sufficiently expressive, the diversity constraint may not
249 provide meaningful improvements and could introduce noise into the selection process.

250 Finally, the approach assumes access to a human labeling oracle and does not explicitly account for
251 labeling errors or annotation costs beyond a fixed budget. In practical applications, labeling quality
252 and cost variability may influence performance and should be considered in future work.

253 Despite these limitations, the results provide strong evidence that combining ambiguity-based selec-
254 tion with diversity-aware sampling improves robustness under heterogeneous distribution shifts.

255 **8 Conclusion**

256 A human-assisted framework for OOD generalization and detection has been explored under het-
257 erogeneous distribution shifts. The results demonstrate that sample selection plays a critical role in
258 improving robustness.

259 Top-k sampling focuses on extreme samples and achieves strong classification performance but fails
260 under complex distribution shifts. AHA improves robustness by focusing on ambiguous samples
261 near the decision boundary. AHA-Diverse extends this by incorporating diversity, leading to im-
262 proved coverage of the input space.

263 Overall, the combination of uncertainty-based selection and diversity-aware sampling provides a
264 balanced tradeoff between accuracy and detection performance. Future work may explore more
265 precise threshold estimation, adaptive selection strategies, and scaling to larger datasets.

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